RMO– 2023

Max Mark – 102 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Let N be the set of all positive integers and $S = \{(a,b,c,d) \in N^4 : a^2 + b^2 + c^2 = d^2\}$. Find the largest positive integer m such that m divides $abcd$ for all $(a,b,c,d) \in S$.
2. Let w be a semicircle with AB as the bounding diameter and let CD is a variable chord of the semicircle of constant length such that C, D lie in the interior of the arc AB. Let E be a point on the diameter AB such that CE and DE are equally inclined to the line AB. Prove that:
   (i) The measure of $\angle CED$ is constant.
   (ii) The circumcircle of triangle CED passes through a fixed point.
3. For any natural number n, expressed in base 10, let s(n) denote the sum of all its digits. Find all natural numbers m and n such that m < n, and $(s(n))^2 = m \quad and \quad (s(m))^2 = n$.
4. Let Ω_1, Ω_2 be two intersecting circles with centres O_1, O_2 respectively. Let l be a line that intersects Ω_1 at A, C and Ω_2 at B, D such that A, B, C, D are collinear in that order. Let perpendicular bisector of AB intersects Ω_1 at points P, Q; and perpendicular bisector of CD intersects Ω_2 at points R, S such that P, R are on the same side of l. Prove that midpoints of PR, QS and O_1O_2 are collinear.
5. Let n>k>1 be positive integers. Determine all positive integers $a_1, a_2, a_3, ..., a_n$ which satisfies
   $$\sum_{i=1}^{n} \sqrt{\frac{k a_i^{\,k}}{(k-1)a_i^{\,k} + 1}} = \sum_{i=1}^{n} a_i = n$$
6. Consider a set of 16 points arranged in a 4 X 4 square grid formation. Prove that if any 7 of these points are coloured blue, then there exists an isosceles right angled triangle whose vertices are all blue.

******
****
**